

Assignment - 4

1) ✓ Fix a vector u in an inner product space (shortly, ips) V . Define a linear functional T on V by $Tv = \langle v, u \rangle$ for $v \in V$.

what is $T^*(\alpha)$ for a scalar α ? [Ans : $T^*(\alpha) = \alpha u$]

2) ✓ Define an operator T on $\mathbb{F}^n$ by $T(a_1, a_2, \dots, a_n) = (0, a_1, \dots, a_{n-1})$.

what is $T^*(a_1, a_2, \dots, a_n)$?

[Ans : $T^*(a_1, a_2, \dots, a_n) = (a_2, \dots, a_{n-1}, 0)$]

3) ✓ prove that a linear operator T on a finite dimensional ips

is invertible iff T^* is invertible.

In that case, show that

$$(T^*)^{-1} = (T^{-1})^*$$

4) Show that on a finite dimensional
ips, a linear operator is unitary
iff it is isometric.

5) Give an example of a
linear operator which is

- a) normal but not self-adjoint
- b) normal but not unitary
- c) unitary but not self-adjoint
- d) self-adjoint but not unitary.

6) Determine a polynomial $q(t) \in R_2[t]$

So that $\int_0^1 p(t) q(t) dt = p(1/2)$ for each $p(t) \in R_2[t]$.

$$[Ans: q(t) = -3/2 + 15t - 15t^2]$$

7) Define $f: \mathbb{C}^3 \rightarrow \mathbb{C}$ by

$$f(a, b, c) = \frac{a + b + c}{3}. \text{ Find a vector } y \in \mathbb{C}^3 \text{ such that}$$

$$f(x) = \langle x, y \rangle \text{ for each}$$

$$x \in \mathbb{C}^3. [Ans: (1/3, 1/3, 1/3)]$$

8) Let T be a linear operator on a finite dimensional ips. Show that if $T^*T = I$, then $TT^* = I$.

Assignment-5

1) ✓ Show that $\left\{ \left(\frac{1+i}{2}, \frac{1-i}{2} \right), \left(\frac{i}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right) \right\}$ is an orthonormal set in $\mathbb{C}^2$.

Express $(2+4i, 2i)$ as a linear combination of the given orthonormal vectors.

2) Let $\{u, v\}$ be an orthonormal set in $\mathbb{C}^{2 \times 1}$. Let $x = (4+i)u + (2-3i)v$. Determine the values of u^*x , v^*x , x^*u , x^*v , and $\|x\|$.

3) Let $\{u, v\}$ be an orthonormal basis for $\mathbb{R}^{2 \times 1}$. Let $x \in \mathbb{R}^{2 \times 1}$ be a unit vector. If $x^t u = \frac{1}{2}$,

then what could be $x^t v$?

4) Let $A \in \mathbb{F}^{m \times n}$. Show the following:

a) For $m \leq n$, $AA^* = I$ iff the rows of A are orthonormal.

b) For $m \geq n$, $A^*A = I$ iff the columns of A are orthonormal.

5) Find a QR-factorization of each of the following matrices.

$$a) \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$b) \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

$$c) \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Assignment-1

1) Let $\{u, v, w\}$ be an orthonormal basis for a subspace of $\mathbb{R}^{5 \times 1}$ and let $x = au + bv + cw$.

If $\|x\| = 5$, $\langle x, u \rangle = 4$ and $x \perp v$, then what are the possible values of a, b , and c ?

2) Let A be the 4×2 matrix whose rows are $[\frac{1}{2}, -\frac{1}{2}]$, $[\frac{1}{2}, \frac{1}{2}]$, $[\frac{1}{2}, -\frac{1}{2}]$ and $[\frac{1}{2}, \frac{1}{2}]$.

a) Determine the projection matrix P that projects vectors onto $R(A)$.

- b) Find an orthonormal basis for $N(A^t)$, and then determine the projection matrix Q that projects vectors onto $N(A^t)$.
- c) Is it true that $P^2 = P^t = P$ and $Q^2 = Q^t = Q$?
- 3) Let y be the orthogonal projection of a vector $x \in \mathbb{R}^{n \times 1}$ onto a subspace V of $\mathbb{R}^{n \times 1}$. Show that $\|y\|^2 = \angle x, y$.
- 4) Find the best approximation of $x \in U$ from V where

$$a) \quad U = \mathbb{R}^3, \quad x = (1, 2, 1),$$

$$V = \text{span} \{ (3, 1, 2), (1, 0, 1) \}$$

$$b) \quad U = \mathbb{R}^3, \quad x = (1, 2, 1),$$

$$V = \{ (\alpha, \beta, \gamma) \in \mathbb{R}^3 \mid \alpha + \beta + \gamma = 0 \}$$

$$c) \quad U = \mathbb{R}^4, \quad x = (1, 0, -1, 1),$$

$$V = \text{span} \left\{ \begin{array}{l} (1, 0, -1, 1), \\ (0, 0, 1, 1) \end{array} \right\}$$

5) Find Least Squares
Solutions of the system $Ax=b$,
where

$$a) A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \\ 2 & -1 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$$

$$b) A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}, b = \begin{bmatrix} 0 \\ 1 \\ -1 \\ -2 \end{bmatrix}$$

b) Find the closest point to $b = (1, 0, 2)$ in the subspace $V = \text{span}\{(1, -1, 1), (1, 2, -1)\}$.